

NEW CHITOSE AP, Japan

WMO: 474250

Lat: 42.775N

Lon: 141.692E

Elev: 25

StdP: 101.03

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.2	-16.0	-21.1	0.6	-17.7	-18.8	0.7	-15.2	12.5	-0.4	11.4	-0.9	1.5	170	0.637

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.6	27.2	22.1	25.9	21.6	24.2	20.7	23.6	26.0	22.6	24.6	21.7	23.6	5.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	17.7	25.3	22.0	16.7	23.9	21.0	15.8	23.0	70.8	26.4	66.8	24.8	63.5	24.3	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	10.1	8.9	DB	-22.3	30.3	2.6	1.5	-24.1	31.5	-25.6	32.4	-27.1	33.2	-28.9	34.4	(4)
(5)			WB	-22.5	25.1	2.6	1.1	-24.4	25.9	-25.9	26.5	-27.3	27.1	-29.2	27.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	-7.0	-6.3	-1.1	5.5	11.1	15.2	19.3	21.0	17.2	10.3	3.4	-3.7	(6)	
	DBStd	10.19	3.77	3.94	3.68	3.12	2.84	2.65	2.58	2.54	3.16	3.38	4.00	4.12	(7)	
	HDD10.0	2152	527	456	344	139	21	1	0	0	0	39	201	426	(8)	
	HDD18.3	4251	785	689	602	385	224	99	19	6	58	251	448	684	(9)	
	CDD10.0	1111	0	0	0	4	55	156	289	340	217	47	3	0	(10)	
	CDD18.3	168	0	0	0	0	0	4	51	88	24	0	0	0	(11)	
	CDH23.3	571	0	0	0	0	3	17	154	319	77	0	0	0	(12)	
	CDH26.7	82	0	0	0	0	1	2	19	55	6	0	0	0	(13)	
(14) Wind	WSAvg	4.1	3.4	3.7	4.3	5.0	5.1	4.6	4.4	4.0	3.8	3.8	3.7	3.3	(14)	
(15) Precipitation	PrecAvg	1233	75	63	57	73	108	91	144	165	166	104	98	90	(15)	
	PrecMax	1569	110	141	101	132	187	255	302	337	277	150	206	168	(16)	
	PrecMin	991	45	25	24	8	40	33	68	73	42	43	60	56	(17)	
	PrecStd	141	15	26	20	35	38	44	54	69	65	29	34	27	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.2	5.2	10.9	18.9	23.0	25.2	28.8	30.2	27.2	21.2	16.1	9.9	(19)	
		MCWB	2.7	2.9	6.2	11.0	14.2	18.5	22.3	24.4	21.6	16.4	12.5	8.5	(20)	
	2%	DB	2.8	3.2	8.0	15.2	20.2	23.0	26.8	28.1	25.2	19.2	14.1	7.1	(21)	
		MCWB	1.2	1.4	4.5	9.2	13.3	17.7	22.1	23.0	20.3	15.2	11.2	5.1	(22)	
	5%	DB	1.1	2.1	6.1	13.1	18.2	21.2	25.0	26.2	24.0	18.0	12.2	4.9	(23)	
		MCWB	-0.4	0.5	3.5	8.1	12.2	16.8	21.1	22.2	20.0	14.3	9.7	3.0	(24)	
	10%	DB	-0.1	1.0	4.8	11.2	16.8	19.9	23.2	25.0	22.8	16.8	10.9	2.9	(25)	
		MCWB	-1.4	-0.6	2.5	6.9	11.8	16.1	20.4	21.6	19.4	13.7	8.3	1.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	3.6	6.8	11.7	15.8	19.9	24.4	25.3	23.6	18.1	13.7	8.9	(27)	
		MCDB	3.7	4.2	8.9	16.5	20.2	23.4	26.9	29.3	25.3	19.9	15.4	9.8	(28)	
	2%	WB	1.1	1.9	5.1	10.4	14.6	18.5	22.8	24.1	22.4	16.4	12.1	5.4	(29)	
		MCDB	2.0	2.9	7.0	13.8	18.3	21.7	25.3	26.9	24.2	18.3	13.8	6.5	(30)	
	5%	WB	-0.1	0.6	3.8	9.2	13.5	17.5	21.9	23.2	21.3	15.3	10.3	3.0	(31)	
		MCDB	0.8	1.8	5.5	12.1	16.9	20.3	24.0	25.2	23.0	17.3	12.0	4.4	(32)	
	10%	WB	-1.7	-0.5	2.6	7.9	12.6	16.6	21.0	22.4	19.9	14.0	8.6	1.0	(33)	
		MCDB	-0.4	0.9	4.4	10.2	15.7	19.0	22.7	24.2	22.0	16.2	10.3	2.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	11.0	8.8	9.7	9.4	7.7	6.4	6.6	9.0	10.6	9.5	9.6	(35)	
		MCDBR	10.1	10.4	9.6	13.0	12.6	10.3	8.9	8.4	9.4	11.1	11.7	10.5	(36)	
	5% WB	MCWBR	9.1	9.1	7.2	7.9	6.9	5.7	4.9	4.5	5.5	7.6	9.5	9.1	(37)	
		MCWBR	9.9	10.1	9.0	11.1	10.3	8.5	7.2	6.9	6.9	9.5	10.8	9.8	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.340	0.391	0.465	0.526	0.544	0.525	0.511	0.480	0.420	0.407	0.388	0.355	(40)	
		taud	2.044	1.897	1.795	1.786	1.837	1.965	2.072	2.176	2.297	2.195	2.143	2.059	(41)	
	Ebn at Noon	Ebn at Noon	753	768	757	748	749	765	771	783	806	757	697	690	(42)	
		Edh at Noon	110	152	192	209	204	180	160	140	115	113	101	100	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.54	2.34	3.50	4.58	5.11	4.96	4.49	4.25	3.69	2.71	1.63	1.24	(44)	
	RadStd	RadStd	0.10	0.17	0.24	0.47	0.44	0.41	0.49	0.32	0.33	0.15	0.14	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page